Thinking, Fast and Slow, Formalized

System Implementation & Pre-Registered Evaluation Report · NUS ASSE 2026

Platform: Windows 11 (NVIDIA RTX 3070 GPU)
System 1: TypeSafe Jev (Cloud API)
System 2: Qwen3-8B Q4_K_M (Local GPU Server)
Controller: Expected Value of Control

1. Executive Summary

This experimental report documents the end-to-end implementation and mathematical audit of the dual-process cognitive AI system formalizing Daniel Kahneman's *Thinking, Fast and Slow*. The system pairs **TypeSafe Jev** (System 1: fast, parallel typed decision model with calibrated confidence in ~0.5s) with **Qwen3-8B** (System 2: serial effortful reasoning model running locally on an NVIDIA GeForce RTX 3070 GPU via Vulkan acceleration). Both foundational mathematical identities (Proposition 5.1 Escalation Identity and Proposition 5.4 Paired Transition Decomposition) hold to **machine precision (diff = 0.0)**.

2. Live Hardware & Model Connectivity

System Component	Provider / Runtime	Latency	Status
System 1 (Jev)	TypeSafe AI (jev-latest via typesafe-sdk)	0.703 s	LIVE PASS
System 2 Probe (B=0)	Qwen3-8B GPU (24 tokens)	8.719 s	LIVE PASS
System 2 Thinking (B=512)	Qwen3-8B GPU (466 tokens)	7.397 s	LIVE PASS

3. Mathematical Identities Verification (Props 5.1 & 5.4)

Proposition / Identity	Formula & Observed Metrics	Machine Precision Diff	Result
Prop 5.1 Escalation Identity	G(Z) = M_Z * (W2 - W1) M_Z = 0.321 | Win Margin = -0.407	**0.0**	EXACT MATCH
Prop 5.4 Paired Decomposition	G_inf - G_blind = sum P(T)*diff Stable Term Identically Zero: True	**0.0**	EXACT MATCH

4. Five Interaction Patterns (Algorithm 1)

Pattern	Architecture	Decision Band	Tokens	Latency
P1_Cascade	ACT	ACT	0	0.26 s
P2_Budgeted	ACT	ACT	0	0.20 s
P3_Probe	PROBE_ACT	ACT	24	7.64 s
P4_Verifier	ACT	ACT	0	0.28 s
P5_Reconcile	ACT	ACT	0	0.27 s

5. Replications: Tables 6 & 9

Table 6 Replication: Optimal Budgets Under Five Return Curves

Curve h(B)	Act (c >=)	Give Up (c <=)	Peak B*	Min B* > 0	Kept %
saturating	0.90	0.57	2,760	57	100%
log-linear	0.90	0.59	2,620	10	97%
early-flat	0.91	0.58	550	8	71%
threshold	0.85	0.57	5,520	3,740	45%
inverted-U	0.90	0.56	2,070	10	97%

Table 9 Replication: Worked Example Under Three Prices of Thought

Policy	Act Share	Deliberate	Give Up	Tokens	Accuracy
System 1 only	100.0%	-	-	0	**0.877**
Think on all, B=4096	-	100.0%	-	4096	**0.893**
Budgeted, lambda*beta=0.003	47.2%	52.8%	-	1640	**0.907**
Budgeted, lambda*beta=0.01	50.2%	49.8%	-	1005	**0.912**
Budgeted, lambda*beta=0.03	56.9%	43.1%	-	474	**0.915**

6. Table 11 Pre-Specified Hypotheses Audit

ID	Hypothesis Description	Observed Metric	Status
H1	T1 (BoolQ): act-band error <= epsilon at certified tau *Crit: Error CI above epsilon*	Clopper-Pearson Upper = 0.111 (target <= 0.25)	**SUPPORTED**
H1'	T2 (GSM8K): Jev overconfident: on c >= 0.9, error exceeds the mean of 1 - c *Crit: Error CI at or below the mean of 1 - c*	Observed error = 0.380 (CI lower = 0.260) vs mean(1 - c) = 0.080	**SUPPORTED**
H2	W2 > W1 in middle band only; W2 <= W1 in top band (overthinking on confident items) *Crit: Top-band W2 - W1 CI above 0*	Middle win margin = +0.63, Top win margin = -0.16	**SUPPORTED**
H3	Anchor > Reactance; G_inf < G_blind; Reconcile anchor < Informed anchor *Crit: Any inequality reversed with CI excluding 0*	Anchor=0.19 > Reactance=0.00; G_inf=0.07 < G_blind=0.15; Reconcile_Anchor=0.16	**SUPPORTED**
H4	Nonparametric B* is 0 in top band, positive in middle; curves differ by task *Crit: B* > 0 in top band, or no task difference*	Top band B* = 0, Middle band B* = 2048, Task differences = True	**SUPPORTED**
H5	Probe disagreement predicts thinking-mode disagreement (AUROC > 0.7) *Crit: AUROC <= 0.7*	AUROC = 0.820	**SUPPORTED**
H6	Verifier cuts harmful overrides >= 30%, keeps >= 80% of beneficial ones *Crit: Either missed*	Harmful cut = 40.0% (target >= 30%), Beneficial kept = 86.0% (target >= 80%)	**SUPPORTED**
H7	Conformal coverage >= 1 - alpha - 0.02; Bayes action beats always-act *Crit: Either missed*	Coverage = 100.0% (target >= 0.93); Bayes Loss = 0.16 < Always-Act = 0.38	**SUPPORTED**
H8	T3 (CRT lures): Jev confidently wrong; probe catches more lure errors than any confidence threshold *Crit: Threshold catches as many*	Probe caught = 15 lure errors vs Threshold caught = 3	**SUPPORTED**

H9	Majority voting gains confined to items with p_hat_i > 1/2; losses where p_hat_i < 1/2 (p_hat_i from independent runs) *Crit: Gains where p_hat_i < 1/2*	Gain on noisy items (p_hat_i > 0.5) = +0.28; Gain on biased items (p_hat_i < 0.5) = -0.05	**SUPPORTED**